

Coexistent Evaluation on the Effects of Radar PRF and 5G Symbol Based Scheduling

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Abstract—The increasing demand for spectrum necessitates the shared use of frequency bands between commercial wireless systems and incumbent radar operations. The 3.5 GHz Citizens Broadband Radio Service (CBRS) band in the United States is a prime example where 5G New Radio (NR) deployments must coexist with federal radar systems. This paper explores the relationship between pulsed sensor interference, particularly its Pulse Repetition Frequency (PRF), and the resulting effects on 5G downlink throughput. We present a comprehensive research approach combining a simulation framework utilizing open-source 5G software stacks and custom radar modeling in MATLAB, validated by over-the-air (OTA) experiments with a commercial UE and Software Defined Radios (SDRs). Our findings demonstrate that 5G and radar coexistence is achievable under certain conditions, and critically, not all radar interference scenarios result in complete blockage of 5G communications. The simulations reveal that for low radar PRFs (below 3 kHz) and under 15% duty cycle, strategic adjustment of 5G symbol-based scheduling can lead to notable improvements in downlink throughput. Furthermore, the results from our real-world OTA testing show good alignment with the simulation outcomes, confirming the viability of our modeling approach. This work identifies specific sensor PRF ranges that are more conducive to cellular coexistence and highlights the potential for preserving 5G performance under high duty cycle conditions by adapting TTI boundaries and symbol allocations.

Index Terms—5G, Radar, Sensor, PRF, PD, Duty Cycle, DL, UL, Scheduler, MAC, MCS, CQI, Spectrum, Co-Existence, Symbol, TTI, Matlab, Model, Harq,srsRAN, open5GS

I. INTRODUCTION

Growing demand for wireless spectrum necessitates new utilization approaches, with spectrum sharing be-

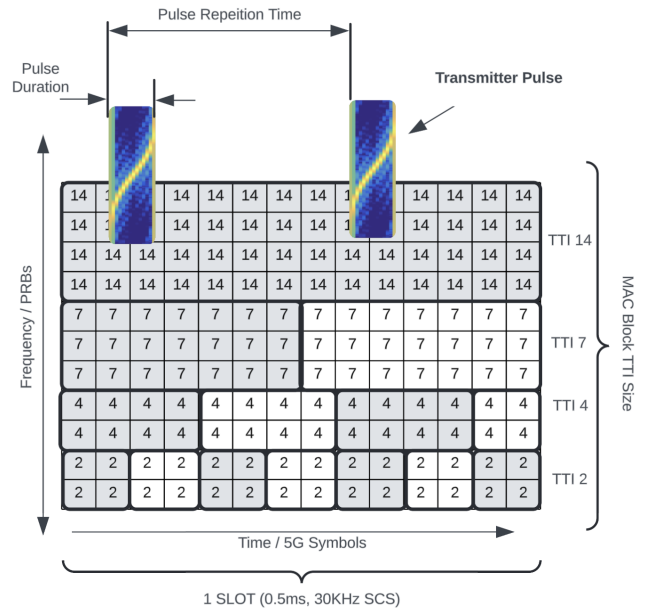


Fig. 1: Example of Radar Pulse intersecting with 5G Transmission Blocks. Blocks can be scheduled in symbol groupings of 2, 4, 7 or the entire slot (14).

ing key. The US FCC's Citizens Broadband Radio Service (CBRS) in the 3.5 GHz band (3550–3700 MHz) exemplifies this need, requiring commercial 5G New Radio (NR) to dynamically coexist with incumbent federal radar systems, posing significant interference challenges [1], [2].

Early coexistence efforts focused on protecting radar from cellular emissions, with NTIA's 2010 report proposing exclusion zones [3]. However, later FCC analyses found radar-to-cellular interference underestimated, prompting further study and mitigation technique development [4].

While much research addresses 5G–radar coexistence, it often focuses on specific radar types and fixed duty cycles (e.g., 0.01, 0.09, 0.1, 1.6, 10, 15, 16, & 20% [3]). Since duty cycle ($DC = PD * PRF$) allows infinite pulse duration (PD) and pulse repetition frequency (PRF) combinations for equal values, studies tied to particular incumbent systems may be too narrow in focus. There’s a clear need to understand how fundamental radar parameters, beyond fixed duty cycles, influence 5G performance.

This paper fills that research gap by investigating pulsed radar interference, specifically how PRF impacts 5G NR throughput. Our approach combines detailed link-level simulations with real over-the-air (OTA) experiments to identify relationships between radar parameters and 5G downlink throughput.

Crucially, we systematically vary PRF while keeping pulse duration constant, spanning a broad range of duty cycles (comparable to NTIA studies and prior work [5]). This method, unlike those focused on specific duty cycles [3], allows us to generalize PRF’s interaction with 5G performance, extending knowledge beyond power-centric views. We also explore leveraging flexible 5G features—such as symbol-level scheduling and dynamic TTI allocation to enhance coexistence under deterministic radar interference.

Through this combined simulation and experimental methodology, our work aims to develop a generalized understanding of radar interference effects, untied to specific incumbent system characteristics [6]. By identifying the sensor signal characteristics most detrimental to 5G, we evaluate adaptive, symbol-based scheduling as a mitigation strategy for interference. Our results indicate that strategic, symbol based TTI scheduling can significantly improve user downlink throughput under radar transmissions. The insights from this research will inform future spectrum-sharing policies and guide the design of both radar and cellular networks for optimized coexistence.

The key contributions of this work are as follows:

- **Generalized Coexistence Framework:** We developed an evaluation framework, combining simulations with over-the-air (OTA) testing, to assess radar–5G coexistence across various radar waveforms and deployment scenarios.
- **Isolation of PRF Effects:** By systematically varying radar pulse repetition frequency (PRF) while holding pulse width constant, we isolated PRF as an independent factor. This approach goes beyond

fixed duty-cycle analyses to reveal how different PRFs uniquely impact 5G link performance.

- **Symbol-Level Scheduling Mitigation:** We investigated and demonstrated 5G symbol-level scheduling as an interference mitigation strategy, evaluating its ability to sustain throughput under pulsed radar interference.
- **Simulation and OTA Validation:** Our findings are validated through comprehensive simulations and controlled OTA experiments, confirming the accuracy of our models and the practical feasibility of our proposed coexistence techniques. We also validated other works results showing user throughput degradation as it relates to radar power level interference.

II. RELATED WORK

Research in coexistence of radar and cellular communications falls under two main thrusts: (1) the impact of cellular transmissions on radar performance and (2) the effect of radar interference on cellular network operations.

A. Cellular Communication Impacts to Radar System:

Radar systems operated by U.S. federal agencies in the 3.3–3.8 GHz range are typically categorized as radio-location or radio-navigation services [7]. For instance, the Navy employs shipborne long-range surveillance radars in the 3550–3650 MHz band, the Army uses land-based battlefield weapon-locating radars in the 3100–3400 MHz range, and the Air Force deploys airborne search and surveillance radars across 3350–3600 MHz [3].

Base station power control was highlighted as a potential mitigation technique to reduce interference to radar in a 2016 study by Raymond [8]. Additionally, LTE transmission power has been shown to correlate directly with increased radar false alarm rates (FAR) [9]. A 2000 report from the NTIA identified numerous pulsed-based radar systems [7] in this frequency band, a consideration that motivates our use of FM pulse-based radar models in MATLAB for experimental simulation.

B. Radar System Interference to Cellular Communications:

Radar interference in OFDM-based networks, particularly on 5G uplink Block Collision Rates (BCR), is well-documented. Radar signal power significantly degrades throughput; simulations confirm rapid 5G

throughput decline with increasing power [9]. While radar PRF and pulse width (PD) also affect error patterns [9], 1800 MHz experiments highlight radar power as the dominant factor in throughput loss, with limited pulse width effect [10]. LTE has been studied to show operations even under low, radar-induced, SNR [11]. Field studies indicate user equipment (UE) is more vulnerable than base stations, and TCP-based FTP traffic suffers more than HTTP [6].

Prior work, including NTIA assessments [3] and LTE studies [11], used 0.1% radar duty cycles to evaluate radar detection degradation. Others have studied LTE FDD, TDD, and Gaussian noise to determine interference-to-noise ratio (I/N) as a key factor in radar detection probability (P_d) [5]. However, such studies, including those on radar-induced BLER/SER in 5G [12], still often focus on specific duty cycles or radar configurations. Additionally, while 5G's impact on radar performance [13], [14] and passive radar sensing via 5G [15] are indirectly related, our work distinctly focuses on comprehensively understanding how radar influences 5G system performance for end-users and enabling robust coexistence.

Building on existing knowledge, our study uses a n78 5G NR testbed with TCP traffic and symbol-level scheduling to investigate how varying radar PRF and power levels specifically influence 5G downlink throughput under realistic conditions. A key differentiator is our systematic exploration of PRF's contribution to radar duty cycle. Stated before, prior works are often limited to specific duty cycles (e.g., 0.1% [5] [7]), we vary PRF with fixed PD to explore a broader range of duty cycles (0% to 25%). By isolating PRF's impact, our approach generalizes beyond specific radar configurations, offering a more comprehensive characterization of the impacts to 5G performance. This understanding extends beyond power-centric views, providing vital insights into PRF's role in 5G interference and informing the design of future coexistent sensors for efficient spectrum sharing.

III. SYSTEM OVERVIEW

To evaluate radar-5G coexistence, we designed a system architecture that enables both simulation-based and OTA-based analysis. Figure 2 illustrates the concept of operations (ConOps), where we model overlapping in-phase and quadrature (IQ) radar interference and analyze its temporal alignment with 5G NR transmissions.

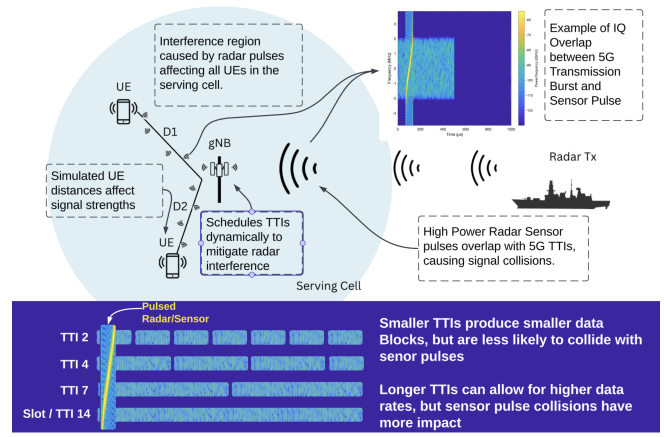


Fig. 2: 5G sensor Con-Ops, showing TTI Overview and IQ interference overlap.

Our framework enables the variation of radar PRF while keeping pulse duration constant. This approach allows us to cover a wide range of duty cycles, from 0% to 25%, extending beyond the specific duty cycles 0.1% explored in prior studies such as [5], and taking a more general approach. By isolating PRF as the key factor contributing to duty cycle, we explore periodic interference more deeply, while also covering a more comprehensive range done by the NTIA[3].

The core idea behind our system is to assess how radar-induced interference aligns with 5G's Transmission Time Interval (TTI), particularly under flexible symbol-based scheduling. By modeling the overlap between radar pulses and 5G symbols at the IQ level, we evaluate the potential for adaptive TTI-based mitigation strategies. The figure shows the radar burst and TTI alignment, providing intuition on when and how throughput may be impacted depending on PRF values.

This system-level framework enables a deeper understanding of radar interference characteristics and helps identify opportunity spaces where dynamic scheduling can mitigate throughput degradation. Our results show that 5G performance can be preserved even under high PRF conditions by intelligently adapting TTI boundaries and symbol allocations.

IV. EXPERIMENT AND SIMULATION CONFIGURATION

Our MATLAB simulation comprises multiple 10-frame scenarios, each configured according to Table I and incorporating IQ level processing. For every UE scheduling allocation, we generate and overlay radar sensor IQ. Subsequent runs increment a unique

TABLE I: Simulation Parameters

Configuration	Values
gNB Configurations	
Scheduler	Round Robin
Modulation and Coding Scheme (MCS)	64 QAM
Max HARQ	16
Sub-Carrier Spacing	30 KHz
Bandwidth	20 Resource Blocks
Slot Pattern	7D1S2U
Slot Pattern Periodicity	5 ms
Sensor Configurations	
Pulse Duration (PD)	50 μ s
PRF	1 - 5000 (Hz)
Duty Cycle Range ($PRF * PD$)	0-25%
Radar Bandwidth	20 MHz
UE Configurations	
Number of UEs	20
Max bps/UE, TTI=2	0.49 (Mbps) 1:20
Max bps/UE, TTI=4	2.08 (Mbps) 1:20
Max bps/UE, TTI=7	3.43 (Mbps) 1:20
Max bps/UE, TTI=14	3.78 (Mbps) 1:20

sensor characteristic, such as the sensor IQ power level (Figure 3) or the PRF (Figure 4). These simulations are executed for four distinct TTI Scheduling allocations.

Our simulation uses a higher HARQ value for increased robustness. However, the simulation has key limitations: control plane signaling is incompletely modeled, and the 5G signal itself consists solely of the Physical Downlink Shared Channel (PDSCH). As a result, the radar interference only affects the PDSCH. By excluding control channel losses from DCI errors or MCS changes, the simulation may appear more robust to interference than a real-world commercial system.

A. Sensor Power level effects

The initial phase of our simulation focused on identifying specific power ratios between the 5G signal and the radar sensor for isolating in later simulations. The effects illustrated in Figure 3 demonstrate phenomena consistent with findings reported in [12] and [9], where the specific power level of the radar signal directly correlates with the throughput experienced by the 5G users. As is evident from the figure, the symbol-based schedulers evaluated exhibit comparable responses to increases in the radar sensor's power level.

B. Sensor PRF Effects

To setup our PRF simulation, we first chose to isolate the radar power level to 20dB greater than the 5G signal. This approach was based on results from Figure

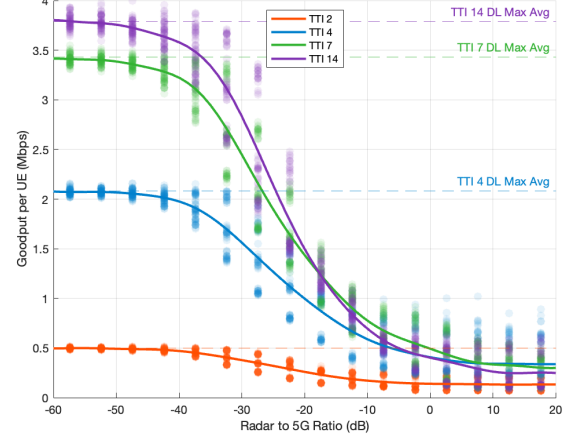


Fig. 3: 20 UEs Simulated throughput against increasing radar sensor power

3, which indicate that beyond a certain power level, increasing the radar's power does not result in further jamming of the signal. We chose "goodput" as our Key Performance Indicator (KPI) rather than "throughput" in our MATLAB model, which includes packet retransmissions. To clarify, Goodput specifically reflects successfully delivered data, and not block retransmission due to HARQ which is labeled as "throughput" in the MAC Layer described by MATLAB.

Our simulation, configured as outlined in Table I, involves generating sensor IQ at the start of each 10-frame run. This sensor IQ is combined with the 5G IQ generated for each user TTI Block throughout the simulation. In each simulation iteration, we increment the Pulse Repetition Frequency (PRF), run the 10-frame simulation again, and distill the results into a single average goodput value. The simulation is run again for the 5G scheduler's TTI (2,4,7,& 14) as shown in Fig. 4.

V. OVER THE AIR DESIGN AND DEMONSTRATION

Our Over-the-Air (OTA) experimental setup is depicted in Figure 5. We emulated the 5G system using a PC running srsRAN as the gNB and open5GS as the 5G Core, with a USRP B200 SDR for radio transmission. A Google Pixel 7 Pro served as the user equipment (UE). For the interference source, we generated custom linear FM pulses in MATLAB and transmitted them with a USRP B200 mini. A critical 10 MHz reference clock synchronized the gNB and radar SDRs to ensure precise timing. The 5G and

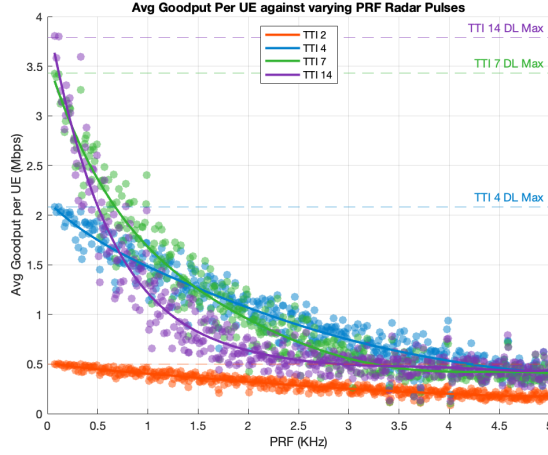


Fig. 4: Simulation Results showing UE throughput under increasing sensor Pulse Repetition Frequencies. Symbol based allocation of TTI 2,4,7 as well as slot (TTI 14)

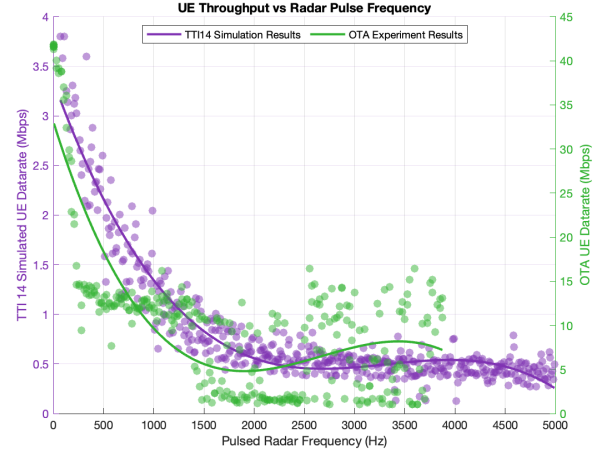


Fig. 6: Experiment Results [RIGHT] w/ 1 UE compared to Simulations [LEFT] avg per UE (1:20)

VI. RESULT ASSESSMENT

Figure 4 shows that deterministic pulsed interference degrades, rather than completely jams, UE performance. While all scheduling types are impacted by the pulse repetition frequency (PRF), each unique TP curve eventually saturates to a non-zero level at 5KHz PRF or 25% Duty Cycle.

The results reveal that performance can be optimized by matching the scheduling type to the PRF. For example, in the 0.5-1.5 kHz PRF range where slot-based scheduling (TTI 14) is heavily impacted, TTI 7 is more effective. At 1 kHz PRF, TTI 7 (1.72 Mbps) improves throughput by 30% over TTI 14 (1.32 Mbps), though this is still a significant reduction from the interference-free maximum of 3.78 Mbps per user.

In the 1.5-4 kHz PRF range, TTI 4 (blue) demonstrates the best overall performance. At 2 kHz, it provides a 17% improvement over TTI 7 and a 90% improvement over TTI 14. In contrast, the shortest allocation, TTI 2 (orange), is the most resilient to PRF changes, but this stability comes at the cost of significantly lower overall throughput.

Figure 6 validates the simulation, showing that our OTA experimental results for slot-based scheduling (TTI 14) follow a similar trend. Both the simulation and OTA tests show a sharp throughput decline at low PRFs ($< 500\text{Hz}$) that saturates at higher PRFs ($> 1.5\text{kHz}$). Crucially, the OTA results confirm that this interference degrades performance without causing total throughput loss or a Radio Link Failure (RLF).

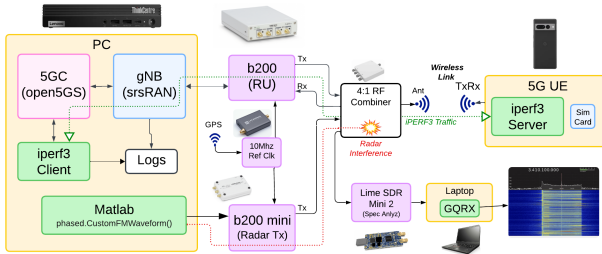


Fig. 5: Over the Air Experiment Setup for generating sensor interference with a commercial UE connected over 5G.

radar signals were then conducted through a 4:1 RF combiner and transmitted OTA to the UE via a small antenna. To capture the combined signal for analysis, a LimeSDR mini was also connected to the combiner, recording I/Q samples using GQRX.

We activated the radar *during* downlink (DL) throughput tests to best match the PDSCH bias of our simulations. The procedure involved initiating UE throughput, activating pulsed sensors for 20 seconds, then deactivating them to observe TP recovery. This was repeated for each incremented Pulse Repetition Frequency (PRF) to determine UE throughput during sensor presence. Results from Figures 6 specifically show comparisons against TTI=14 (Purple), due to the slot based scheduling of srsRAN.

VII. CONCLUSION AND FUTURE WORK

This work presents a systematic investigation into radar–5G coexistence in the CBRS band, focusing on the impact of radar Pulse Repetition Frequency (PRF) and the effectiveness of 5G symbol-level scheduling. Using both simulation and over-the-air (OTA) experimentation, we demonstrate that interference from deterministic pulsed sensors do not completely degrade UE performance, and that certain PRF and scheduling combinations can preserve significant throughput. These findings validate the feasibility of flexible, PRF-aware 5G scheduling as a mitigation strategy.

Our study represents the first stage of a broader, two-stage cognitive-inspired framework for spectrum coexistence. The current stage reflects a “slow-thinking” approach, emphasizing deliberate and interpretable radar and scheduling parameter design based on deterministic modeling. This provides a robust foundation for a “fast-thinking” second stage, which will incorporate adaptive, real-time response strategies to dynamic interference patterns.

Future work will deepen this two-stage cognitive strategy’s *deliberate design* and *adaptive response*. This includes incorporating Radio Link Failure (RLF) and PRACH access failures for broader interference analysis. We will further explore equivalent duty cycles (via PRF and pulse duration) to generalize interference behavior and evaluate the impact of center frequency alignment and bandwidth overlap on coexistence. Crucially, we plan to use machine learning and rule-based methods to infer radar characteristics (PRF, PD, duty cycle) from gNB KPIs (BLER, uplink SNR), facilitating real-time adaptive scheduling. Other 5G NR structures, such as PSS/SSS periodicity, DMRS, CSI-RS, and PUSCH allocation, will also be examined beyond TTI and Max HARQ.

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